

FFP-300K: Scaling First-Frame Propagation for Generalizable Video Editing

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[ffp-300k.github.io](https://github.com/ffp-300k)

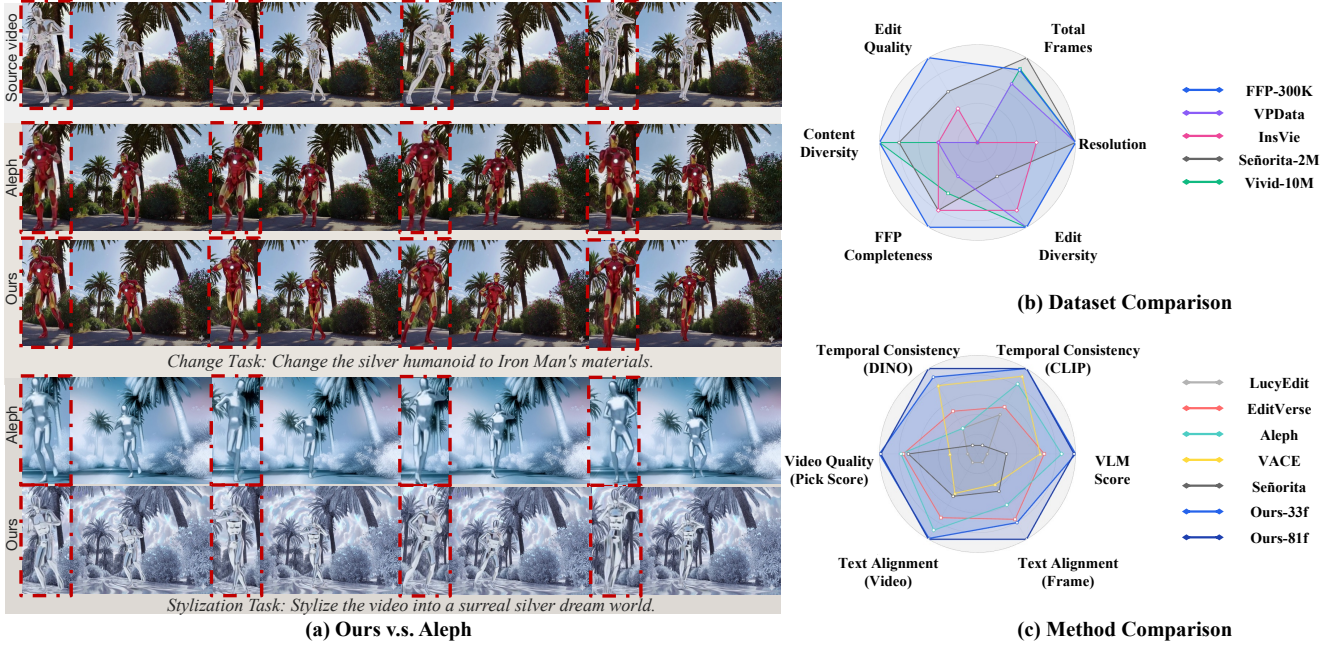


Figure 1. (a) Results from our framework and Aleph [1], a commercial video editing model, with zoom-ins highlighting the main subject. While generally capable, Aleph may **fail to follow the original motion** (top “Change” task) or **present limited visual quality** (bottom “Stylization” task), reflecting the capacity limits of current models. In comparison, based on a first frame edited by Qwen-Edit [31], our framework achieves temporally consistent and visually realistic results on both tasks. For clarity, only Change and Stylization results are shown here; please see the supplementary material for more examples. (b) Overall comparison between our proposed FFP-300K and previous video editing datasets. Each axis represents a key dataset aspect, including total frames for scale, resolution level, supported edit types, completeness of paired source–target data, content diversity across visual content and orientation types, and visual quality of generated target videos, providing an overall assessment of dataset scale, diversity, and consistency. Our FFP-300K is well suited for FFP-based video editing with higher-quality data. (c) Overall comparison between our framework and previous video editing methods, in which ours is generally better among all metrics.

Abstract

First-Frame Propagation (FFP) offers a promising paradigm for controllable video editing, but existing methods are hampered by a reliance on cumbersome run-time guidance. We identify the root cause of this limitation as the inadequacy of current training datasets, which are often too short, low-resolution, and lack the task diversity

required to teach robust temporal priors. To address this foundational data gap, we first introduce FFP-300K, a new large-scale dataset comprising 300K high-fidelity video pairs at 720p resolution and 81 frames in length, constructed via a principled two-track pipeline for diverse local and global edits. Building on this dataset, we propose a novel framework designed for true guidance-free FFP that resolves the critical tension between maintaining first-

frame appearance and preserving source video motion. Architecturally, we introduce Adaptive Spatio-Temporal RoPE (AST-RoPE), which dynamically remaps positional encodings to disentangle appearance and motion references. At the objective level, we employ a self-distillation strategy where an identity propagation task acts as a powerful regularizer, ensuring long-term temporal stability and preventing semantic drift. Comprehensive experiments on the EditVerseBench benchmark demonstrate that our method significantly outperforming existing academic and commercial models by receiving about 0.2 PickScore and 0.3 VLM score improvement against these competitors.

1. Introduction

High-fidelity video editing is a pivotal task with applications spanning professional film production, interactive entertainment, and the surge of user-generated content. An ideal model must provide users with precise control over edits while ensuring realism and temporal coherence. Current diffusion-based methods [13, 21, 26, 35] largely follow two paradigms. The *Instruction-based* approaches [5, 25], while powerful for images, face compounded difficulty in the video domain. A model must simultaneously interpret a user’s textual intent and apply it coherently across a temporal sequence, a dual challenge that often yields results that lag behind the fidelity of their image-based counterparts. In contrast, the *First-Frame Propagation (FFP)* paradigm [15, 16, 19] offers a more pragmatic and powerful alternative by strategically decoupling the editing process. It allows users to leverage the sophisticated and mature ecosystem of image editing tools—from professional software to advanced generative models—to perfect a single frame with high precision. This approach alleviates the model’s burden of semantic interpretation, transforming the complex task of text-to-video editing into a more constrained and well-defined problem: robust temporal propagation. However, this elegant promise of control is undermined by current models’ reliance on cumbersome run-time guidance, such as per-video LoRA fine-tuning [20] or auxiliary inputs like depth maps [15], which incur high computational costs and limit generalization.

This reliance on guidance is not a flaw in the FFP paradigm itself, but a symptom of inadequate training data. Lacking long, high-resolution, and diverse examples, models fail to learn robust temporal priors and are forced to use external guidance as a crutch. This data gap manifests in key limitations: (1) **Insufficient Length and Resolution:** Datasets like Señorita-2M [41] and InsViE [32] feature short, low-resolution clips, hindering the learning of long-range motion and fine details. (2) **Limited Task Diversity:** Many datasets focus on narrow tasks like inpainting (VPData [3]) or fail to distinguish between local and

global edits. (3) **Inconsistent Temporal Alignment:** Hybrid datasets like VIVID-10M [9] mix images and videos, disrupting the learning of continuous motion priors.

To overcome these fundamental limitations, we introduce a synergistic solution comprising a new dataset and a novel framework. First, we present **FFP-300K**, a large-scale dataset engineered to directly address the aforementioned data challenges, which is constructed with a two-track synthesis pipeline. This pipeline leverages a motion-aware generative prior learned by VACE [11] as its backbone to ensure temporal stability, employing mask-based manipulation for precise local edits and depth-guided conditioning for geometry-aware global stylization. This structured approach ensures task diversity and high fidelity. Benefited from the modularized pipeline, our dataset can be easily scaled up to provide sufficient generalization ability, which contains about **290,441** original/edited video pairs at **720p** resolution and a length of **81** frames, providing a rich and diverse foundation for training the next generation of video editing models.

Building upon FFP-300K, we then advance the FFP paradigm by proposing a new framework dubbed **FreeP-rop**, aiming to tackle the core challenge of *balance between referencing the first frame for appearance and referencing the source video for motion* with two key contributions. For architectural level, we design an Adaptive Spatio-Temporal RoPE (AST-RoPE) that creates a content-aware geometry for the model. It learns from the source video to dynamically remap the spatio-temporal geometry, effectively disentangling the two references: it reduces the positional “distance” to the first frame to anchor appearance, while simultaneously rescaling the temporal axis to match the source video’s motion. As for the objective level, we introduce a self-distillation strategy, in which the virtually created identity propagation task acts as a powerful regularizer, ensuring that the relational structure between the edited first frame and all subsequent frames follows a stable trajectory. This prevents semantic drift and ensures the edit’s influence remains potent throughout the video.

We comprehensively evaluate our framework on the EditVerseBench [12] benchmark, demonstrating superior performance against recent models including both academic ones such as EditVerse [12] and commercial ones such as Aleph [1] in both visual fidelity and temporal coherence. Our main contributions are:

- We introduce **FFP-300K**, a large-scale dataset for FFP-based video editing, and the principled two-track generation pipeline used for its creation, addressing key limitations in prior data.
- We propose the novel Adaptive Spatio-Temporal RoPE (AST-RoPE) which disentangles appearance and motion.
- We introduce a powerful self-distillation strategy, which is crucial for maintaining the temporal stability and visual

integrity required for guidance-free generation.

2. Related Work

Instruction-Based Video Editing Models. Instruction-based methods edit videos by interpreting natural language prompts. This paradigm is broadly divided into inversion-based and inversion-free approaches. Inversion-based models like VideoSwap [7] and VideoDirector [30] first map a source video into a latent noise space for editing. While this can yield precise results, the inversion process introduces significant computational overhead, limiting practical application. To circumvent this, inversion-free models are trained on large-scale datasets to generalize across diverse editing instructions. For instance, InsV2V [5] adapts image-to-image translation principles to video, while LucyEdit [25] and EditVerse [12] introduce architectures to better integrate textual and visual conditioning. However, due to the intrinsic difficulty of this task, current instruction-based methods fall far behind their image counterparts.

FFP-Based Video Editing Models. The FFP paradigm offers a more controllable alternative by decomposing video editing into two steps: user-driven first-frame modification and automated temporal propagation. Early methods like AnyV2V [14] and Videoshop [6] demonstrated the potential of this approach but struggled with complex motion. Subsequent works sought to improve temporal coherence but introduced significant dependencies. For example, I2VEdit [19] requires costly per-video fine-tuning, rendering it unscalable. Others, like StableV2V [15] and GenProp [16], rely on auxiliary guidance such as depth maps, optical flow, or predicted masks to preserve structure. Such reliance on external guidance complicates the pipeline and limits model generality due to dependence on auxiliary input quality. Our approach, by contrast, enables fully guidance-free propagation, i.e. conditioning solely on the source video and edited first frame to achieve temporally coherent and controllable results.

Video Editing Datasets. The capabilities of video editing models are fundamentally shaped by the data they are trained on. Several large-scale datasets have been introduced to advance the field. Datasets like EffiVED [40] and VPLM [38] pioneered synthetic data generation for instruction-based tasks, while Señorita-2M [41], VIVID-10M [9], VPData [3], and InsViE [32] significantly increased the scale and diversity of available data for object-level editing. Others such as IVEBench [4] mainly focus on evaluation. However, existing datasets limit robust FFP model development with low-resolution, short clips and unclear distinctions between local and global edits. This forces models to rely on brittle, short-range priors, requiring the external guidance our method eliminates. Our FFP-300K dataset overcomes these issues with high-resolution (720p), long-form (81-frame) videos and separate tracks for

local and global editing, establishing a standardized training set for generalizable FFP models.

3. Scalable FFP Data Construction Pipeline

To address the need for a large-scale, high-fidelity dataset for FFP research, we construct **FFP-300K**. Our data generation framework is a **two-track modular pipeline** designed to produce semantically aligned video editing pairs at 720p resolution. Unlike unified pipelines, our framework operates via two independent and specialized branches to maximize quality for distinct editing categories:

- (1) **Local Editing:** Built upon the Koala-36M [28], this track focuses on fine-grained, object-level operations such as swapping and removal.
- (2) **Global Stylization:** Derived from the Omni-Style [29], this track emphasizes full-scene stylization.

Each branch employs a tailored process of perception, captioning, and synthesis, culminating in a standardized dataset that supports both instruction-based and First-Frame Propagation (FFP) video editing frameworks.

3.1. Local Editing

The local editing branch generates precise object-level modifications. The process integrates large vision-language models (VLMs) for reasoning, advanced segmentation models for spatial localization, and a powerful video inpainting model for synthesis.

Automated Editing Pipeline. For each source video from Koala-36M, we first use Qwen2.5-VL-72B-Instruct [27] to analyze the first frame and identify primary editable objects. Subsequently, Grounded-SAM2 [23] performs instance segmentation to produce frame-wise mask videos, providing precise spatial constraints. These masks, along with task-specific captions, guide the video inpainting model VACE [11] to synthesize the edit.

- **For Swap tasks**, the original caption is used to guide VACE in replacing the masked object while preserving the background context.
- **For Removal tasks**, we prompt Qwen2.5-VL to generate a modified caption that explicitly describes the scene without the target object (e.g., "a street with a bench" instead of "a street with a person sitting on a bench"). This caption then guides VACE to remove the object and plausibly reconstruct the background.

Refining Edits with Mask and Bounding Box Strategies. To optimize visual consistency, we discovered that the nature of the spatial conditioning is critical. We employ a **mask erosion strategy** to preserve only the boundary regions of the target mask, encouraging VACE to better leverage its internal priors for coherent inpainting. Furthermore, we experimented with two complementary conditioning schemes: providing VACE with only the eroded mask

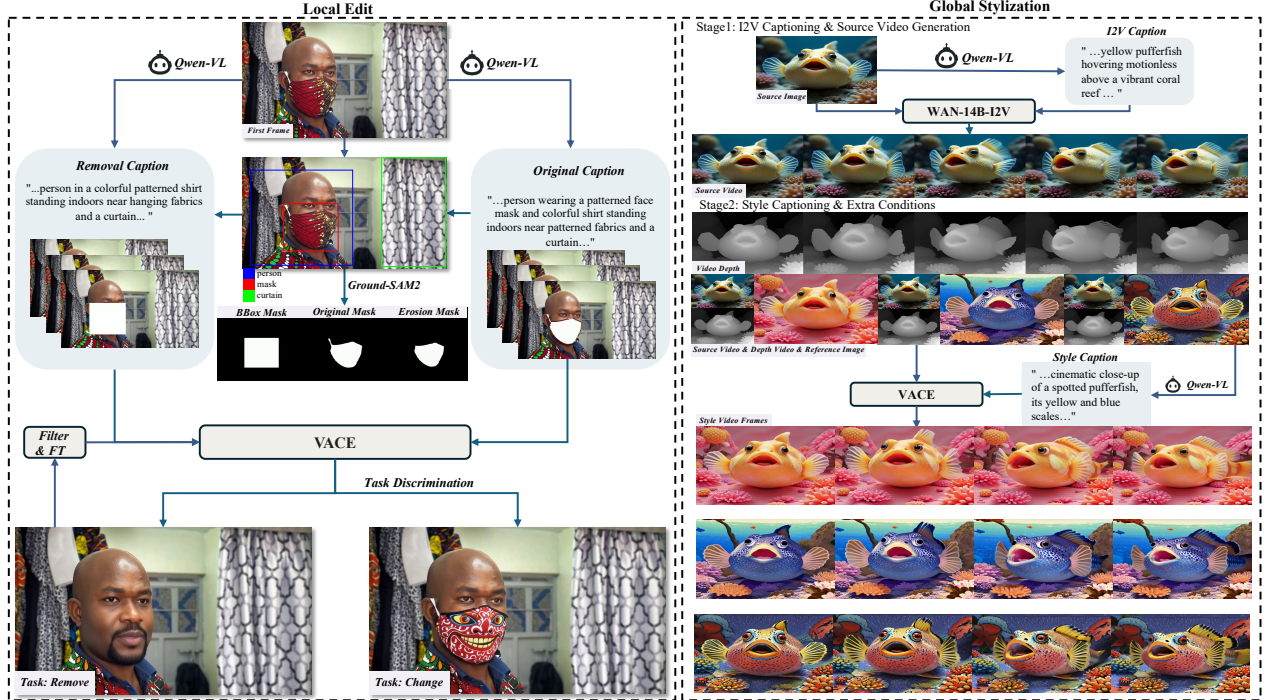


Figure 2. **Overview of our Data Construction Pipeline.** Our pipeline has two parallel tracks. **Left: The local editing track** performs object Swap and Removal. For swapping, we use target objects and captions from the source video to generate edits with erosion masks, followed by a quality filtering step. For removal, captions are constructed and paired with bounding-box masks to generate the edited videos. Notably, filtered samples are used to refine our VACE [11] model, which then regenerates the entire removal subset for higher quality (Sec. 3.1). **Right: The global stylization track** first generates source videos from images using Wan-I2V. It then combines these source videos, style reference images, and corresponding depth videos to produce high-fidelity stylized results (Sec. 3.2).

(*without-bbox*) versus providing both the mask and the object’s bounding box (*with-bbox*). Our empirical analysis revealed a clear task-specific preference:

- **Swap tasks** benefit from the *without-bbox* approach, as the lack of a hard spatial constraint prevents artifacting and yields more semantically natural object integration.
- **Removal tasks** are more successful with the *with-bbox* configuration, which provides a strong spatial prior that ensures complete object erasure and consistent background reconstruction.

This insight informs our quality control process, where we generate both high-quality variants.

3.2. Global Stylization

The global stylization branch transforms the entire visual appearance of a scene. Built upon the diverse Omni-Style dataset [29], this track uses a two-stage process to ensure both semantic coherence and high stylistic fidelity.

Stage 1: Source Video Generation. We first use Qwen2.5-VL to analyze each artistic image from Omni-Style and generate a cinematic video caption describing its scene, atmosphere, and tone. This caption is then used to prompt the Wan2.1-14B-I2V [26] to synthesize a source video, ensuring the generated motion and content are semantically aligned with the reference style image.

Stage 2: Stylized Video Generation. Next, Qwen2.5-VL generates a detailed style caption by observing both the reference style image and the synthesized source video. This caption, which describes color palettes and textures, guides VACE in the stylization process. To preserve geometric structure, we provide VACE with depth maps extracted by Video Depth Anything [34]. This combination of semantic guidance (style caption), structural guidance (depth), and appearance reference (style image) allows VACE to generate the final, temporally coherent stylized video.

3.3. Quality Control and Curation

A multi-stage filtering and verification process is applied to ensure the final dataset’s quality and semantic integrity.

Iterative Refinement for Removal Tasks. The removal subset underwent a particularly rigorous curation loop to maximize precision. First, Qwen2.5-VL automatically screens all generated videos for the removal task to filter out low-fidelity pairs, resulting in an initial set of nearly 40,000 candidates. This was followed by manual verification, yielding 14,389 high-quality samples. We then used this curated set to fine-tune the VACE model, significantly enhancing its removal capabilities. Finally, this improved VACE model was used to regenerate the entire removal subset, achieving cleaner background restoration.

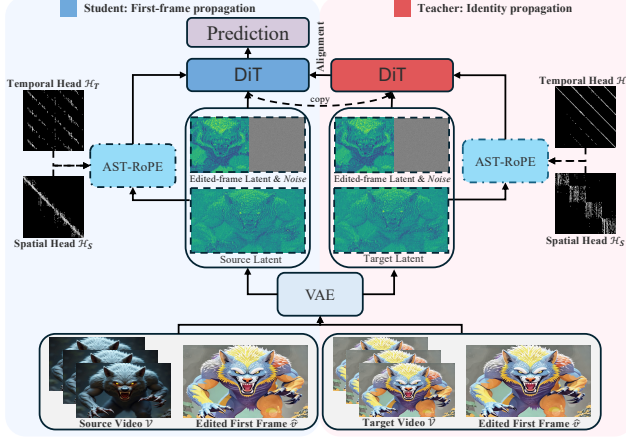


Figure 3. **Overview of training paradigm.** Left: The source video and edited frame are encoded. The source latent informs our AST-RoPE module for adaptive spatio-temporal scaling. Right: The target video is processed identically to extract a latent DiT embedding, which is used to align the generation process.

Final Verification and Statistics. All generated videos (swap, removal, and stylization) undergo a final semantic verification by Qwen2.5-VL to ensure a precise correspondence between the edit instruction and the visual transformation. After filtering and deduplication, our final FFP-300K dataset comprises 290,441 high-quality video pairs (source/edited video). This includes 143,913 stylization, 40,000 removal, and 106,528 swap/modification tasks. All videos are standardized to 720p resolution and 81 frames, making FFP-300K a robust and large-scale resource for advancing video editing research.

4. Methodology

Our proposed FFP framework is designed to intrinsically handle temporal consistency, removing the need for explicit run-time conditions. The model is built upon a powerful conditional video model, which we adapt for the FFP task. Two core innovations, i.e. an adaptive positional encoding scheme and a self-distillation training objective, are introduced to resolve the core tension between appearance propagation and motion fidelity.

4.1. Preliminary

Problem Formulation. Our goal is the First-Frame Propagation (FFP) task: given a source video $\mathcal{V} \in \mathbb{R}^{F \times H \times W \times 3}$, where F, H, W denotes number of frames, height and width respectively, and an edited first frame $\hat{v} \in \mathbb{R}^{H \times W \times 3}$, we aim to generate a target video $\hat{\mathcal{V}}$ that preserves the motion of $\mathcal{V}$ while propagating the edit from $\hat{v}$.

Adapting Fun-Control for FFP. Our method is built upon Fun-Control, a powerful conditional video generation model derived from Wan 2.1 [26]. Fun-Control is designed to aggregate conditioning information from both a video

and a reference image, providing a strong prior for learning motion from the conditioning video while inheriting appearance from the conditioning image. This design is naturally suited for FFP, but was not originally designed for our specific task, presenting two key limitations:

1. Its conditioning videos are low-level signals (e.g., depth maps), not the full RGB videos required in our case.
2. Its reference images are often spatially unaligned, whereas in FFP, $\hat{v}$ is aligned in most regions.

Therefore, task-specific adaptations are necessary. Formally, given $\mathcal{V}$ and $\hat{v}$, we first extract their corresponding VAE latents: $z_{src} \in \mathbb{R}^{F' \times H' \times W' \times C}$ and $\hat{z} \in \mathbb{R}^{H' \times W' \times C}$, where C denotes feature channels. The first-frame latent $\hat{z}$ is then padded with zeros along the temporal dimension and concatenated with the noisy latent z , the source latent z_{src} , and a binary mask $M \in \mathbb{R}^{F' \times H' \times W'}$ (indicating the first frame) along the channel dimension. The resulting composite latent is fed into the DiT backbone for velocity prediction. By fine-tuning this model on our FFP-300K dataset using a flow matching objective, it effectively adapts its pre-trained motion prior to the specific requirements of FFP-based video editing.

4.2. Adaptive Spatio-Temporal RoPE

The self-attention mechanism in a Diffusion Transformer (DiT) relies on Rotary Position Embeddings (RoPE) [24] to understand spatio-temporal relationships. However, standard RoPE imposes a static coordinate system that is ill-suited for FFP. Its uniform temporal progression is agnostic to the source video’s intrinsic motion, and its fixed spatial distances hinder the propagation of the edited first frame, which must serve as a global content anchor.

To overcome this, we introduce **Adaptive Spatio-Temporal RoPE (AST-RoPE)**, a mechanism that endows the DiT with the ability to dynamically adapt its understanding of space and time based on the source video’s content. Instead of a static grid, AST-RoPE learns to modulate the perceived positions of tokens, guiding self-attention to generate motion and appearance that is faithful to the source. This is achieved by predicting content-aware scaling coefficients that separately adjust the RoPE for specialized spatial and temporal self-attention heads.

Source-Aware Scaling Coefficient Prediction. Inspired by the observation of head specialization in DiTs [18, 33], we classify the attention heads in each layer into a static set of Spatial Heads ($\mathcal{H}_S$) and Temporal Heads ($\mathcal{H}_T$). For each video, a lightweight transformer module followed by a two-head MLP predicts a spatial scaling factor α_S and a temporal scaling factor α_T directly from the source latent z_{src} . This allows the model to infer high-level properties from the source video. For instance, predicting a smaller temporal scaling factor for a video with rapid motion.

We apply these coefficients distinctly to each head set.

For spatial heads ($\mathcal{H}_S$), to enhance the first frame’s influence, we use α_S to modulate its perceived positional distance. Specifically, the first item of temporal indice is offset from 0 to $\alpha_S \cdot F'$. By learning to predict $\alpha_S < 1$, we reduce the effective distance between the first frame and all other frames, especially the ending ones. This biases self-attention to assign higher scores between tokens in the edited first frame and those in subsequent frames, ensuring its content is robustly propagated.

For temporal heads ($\mathcal{H}_T$), we use α_T to rescale the temporal axis for all frames. The original temporal indices $[0, 1, \dots, F-1]$ are transformed to $[0, \alpha_T, \dots, \alpha_T(F-1)]$. This operation effectively stretches or compresses the temporal manifold. For a source video with rapid motion, the model can learn a smaller α_T , reducing the perceived temporal distance between frames and encouraging the temporal heads to model more intense motion.

4.3. Self-Distillation with Identity Propagation

To enforce precise motion dynamics and first-frame reference, which standard flow matching fails to sufficiently constrain, we introduce a self-distillation paradigm. Our key insight is that the model’s own internal processing of the source video provides the ideal alignment target. We implement this via a parallel **identity propagation** task, where the “teacher” task is to reconstruct the ground-truth target video $\hat{\mathcal{V}}$ from itself, i.e. conditioned on the $\hat{\mathcal{V}}$ and its first frame $\hat{v}$. This identity mapping forces its internal latents to perfectly encode the desired spatio-temporal dynamics. We then use distillation losses to align the standard “student” FFP task’s representations with this idealized “teacher” representation, ensuring faithful motion preservation.

Inter-Frame Relational Distillation. To ensure global motion patterns are preserved, we distill the frame-to-frame similarity structure, inspired by VideoREPA [39]. Given a latent representation $z^l \in \mathbb{R}^{F' \times H' \times W' \times C}$ from the l -th DiT block of the FFP task, and the corresponding latent $\hat{z}^l$ from the identity propagation task, we first downsample them spatially by a factor of K_S to focus on motion over appearance. Let the resulting latents be z_{ds}^l and $\hat{z}_{ds}^l$, with $N = (H'W')/K_S^2$ spatial tokens. Based on these two latents, the motion alignment can be calculate as:

$$G = \text{Gram}(z_{ds}^l) \quad (1)$$

$$\hat{G} = \text{Gram}(\hat{z}_{ds}^l) \in \mathbb{R}^{F' \times N \times F' \times N} \quad (2)$$

$$\mathcal{L}_{motion} = \frac{1}{F'(F'-1)} \sum_{i,j=1}^{F'} \sum_{i \neq j} |G_{i,:,j,:} - \hat{G}_{i,:,j,:}| \quad (3)$$

where Gram denotes gram matrix along the channel dimension. $\mathcal{L}_{motion}$ minimizes the distance between the inter-frame relationships of the FFP latent and the identity propagation latent, which are indicative of motion, remain consistent with the source video’s dynamics.

First-Frame Consistency Loss. While motion alignment captures global structure, we need a focused mechanism to ensure the edit from the first frame propagates its influence consistently. We propose a novel loss based on Maximum Mean Discrepancy (MMD) to align the evolution of token-wise relationships with respect to the first frame.

For a given frame i , we compute the token-wise similarity matrix between the first frame and frame i : $S_i = z_1^l (z_i^l)^T \in \mathbb{R}^{N \times N}$, where $z_1^l, z_i^l \in \mathbb{R}^{N \times C}$ are the (down-sampled and reshaped) latents for the respective frames. Each of the N rows of S_i is a feature vector describing how a token in the first frame relates to all tokens in frame i . We treat this set of N row vectors as an empirical distribution P_i over an N -dimensional relation space.

We then use MMD with RBF kernel $k(\cdot, \cdot)$ to measure the divergence between the relational distribution of frame i and that of the first frame (an identity relation), yielding a temporal drift score $d_i = \text{MMD}^2(P_1, P_i)$, along with the identity propagation counterpart $\hat{d}_i$, which are constrained to ensure similar evolution with each other:

$$\mathcal{L}_{\text{MMD}} = \sum_{i=2}^F |d_i - \hat{d}_i| \quad (4)$$

This loss regulates that the propagation of the first-frame edit follows a natural dynamic trajectory, as learned from the idealized identity task, preventing the edit’s influence from fading or becoming distorted over time.

Overall Training Objective. Our final training objective combines the standard flow matching loss $\mathcal{L}_{\text{FM}}$ with our two proposed distillation objectives:

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda_{\text{motion}} \mathcal{L}_{\text{motion}} + \lambda_{\text{MMD}} \mathcal{L}_{\text{MMD}}, \quad (5)$$

where λ_{motion} and λ_{MMD} are hyperparameters. Unlike methods that distill from external, generalist models [39], our self-referential guidance is uniquely suited to FFP, as it distills from a teacher that has perfect knowledge of the source video’s specific motion, ensuring edits are propagated without corrupting its essential temporal character.

5. Experiments and Results

5.1. Experiment Setup

Implementation Details. We finetune Fun-Control using LoRA [8] for 2 epoches with rank is set to 128. AdamW [17] is utilized for training with a learning rate of 2×10^{-4} and cosine decay. λ_{motion} and λ_{MMD} are set to 5 and 1. For fair comparison with previous methods, for the main experiments we train two variants of our model with 81-frame videos and 33-frame videos. For ablation study, the one trained with 81-frame videos is engaged in.

Benchmark and Metrics. For evaluation, we adopt EditVerseBench [12], a comprehensive benchmark for video

Type	Method	Resolution	Frames	Temporal Consistency		Text Alignment		Video Quality	VLM Evaluation
				CLIP $\uparrow$	DINO $\uparrow$	Frame $\uparrow$	Video $\uparrow$	Pick Score $\uparrow$	VLM Score $\uparrow$
Training-free	TokenFlow [22]	640×336	48	0.987	0.989	26.779	24.244	20.058	5.067
	STDF [36]	576×320	24	0.965	0.964	26.422	23.768	19.817	4.911
Instruction-based	InsV2V [5]	384×384	32	0.972	0.969	25.923	23.092	19.611	5.252
	LucyEdit [25]	832×480	81	0.985	0.984	26.398	23.491	19.611	5.678
	EditVerse [12]	624×352	64	0.986	0.986	27.776	25.293	20.132	7.104
	Aleph [1]	1280×720	64	0.989	0.984	28.087	24.837	20.291	7.154
FFP-based	VACE [11]	832×480	61	0.990	0.989	27.169	24.188	20.095	6.072
	Señorita [41]	864×448	33	0.981	0.982	27.243	24.404	19.786	6.991
	Señorita [41]*	864×448	33	0.989	0.987	27.754	24.657	19.913	7.341
	Ours-33f	1280×720	33	0.991	<u>0.990</u>	<u>28.293</u>	<u>25.398</u>	20.419	7.631
	Ours-81f	1280×720	81	0.991	0.991	28.316	25.925	<u>20.405</u>	<u>7.600</u>

Table 1. **Quantitative comparison.** We compared three types of video editing methods on EditVerseBench. The best results are highlighted in **bold**, and the second-best results are underlined. As shown, our **33f** and **81f** variants achieve the best performance across all automated evaluation metrics, establishing state-of-the-art results on EditVerseBench. Señorita* refers to using Qwen-Edit [31] to edit the first frame.

editing that covers 20 diverse editing categories. Since our method focuses on FFP-based video editing, we further filter the benchmark to 125 videos with stable temporal structures that can be evaluated under a propagation setting. Then Qwen-Edit [31] is leveraged to generate the edited first frame for these videos. We follow the six metrics defined in EditVerseBench: VLM editing quality, PickScore, Frame score, Video score, CLIP text–image alignment, and DINO-based temporal consistency. To better assess long-sequence propagation, we extend the VLM evaluation from 2 frames to 10 sampled frames. Different from the original setup that uses GPT-4o [10] as the evaluation model, we replace it with Qwen2.5-VL-72B-Instruct [2] to ensure full reproducibility and consistency across evaluation runs.

Competitors. We directly adopt the baseline models provided in EditVerseBench[12], including Token-Flow [22], STDF [36], InsV2V [5], Lucy-Edit [25], Señorita-2M [41], and Aleph [1], to ensure a fair and consistent comparison.

5.2. Quantitative Comparison

In Tab. 1 we present the quantitative comparison between the competitors and two variants of our method. For fair comparison, we test Señorita with the same edited first frames as ours. Our method, in both 33-frame (**Ours-33f**) and 81-frame (**Ours-81f**) configurations, consistently outperforms all competing approaches across the board. Specifically, Ours-81f achieves the highest scores in temporal consistency (0.991 CLIP score, 0.991 DINO score) and video-level text alignment (25.925), showcasing its exceptional ability to maintain coherence over longer sequences. Furthermore, Ours-33f obtains the top scores in perceptual quality (20.419 Pick Score) and semantic correctness (7.631 VLM Score), indicating superior alignment with user intent. Notably, our model surpasses not only other FFP-based methods like VACE [11] but also strong instruction-based models, including the commercially used Aleph [1]. This highlights the effectiveness of our approach in achieving a superior balance of temporal stability, edit fidelity, and

overall visual quality.

5.3. Qualitative Comparison

We further provide qualitative results to visually demonstrate the advantages of our framework. As illustrated in Fig. 4, previous instruction-based methods such as Aleph and EditVerse mainly suffer from the problem of unsuitable edited first frame, such as the wrong position of starfish in Fig. 4(a), and failure to preserve the content in the original videos. Moreover, it is noteworthy that videos generated by EditVerse also has the flickering problem, which cannot be fully presented with static frames but will be shown in the supplementary material. On the other hand, Señorita, as a FFP-based method, is limited with the video quality, showing mosaic in the bottom of each frame. In contrast, our method produces results with not only longer duration, but also significantly better general quality, accurately preserving object structure and scene layout while maintaining global temporal consistency. This qualitative superiority verifies that the combination of our proposed techniques for consistency modeling and curated high-fidelity dataset enables robust editing propagation across diverse and challenging real-world scenarios.

5.4. User Study

To further evaluate perceptual quality and editing accuracy, we conducted a user study where participants rated videos on a 1–5 scale based on: (1) **Editing Accuracy (EA)**: instruction adherence and semantic consistency, (2) **Motion Accuracy (MA)**: motion fidelity to the source video, and (3) **Video Quality (VQ)**: temporal smoothness and realism. With 15 participants each assessing 8 random videos from EditVerseBench, our method achieved the highest mean scores in all criteria (Tab. 2), demonstrating user preference for its precise alignment and stable dynamics, consistent with quantitative results.

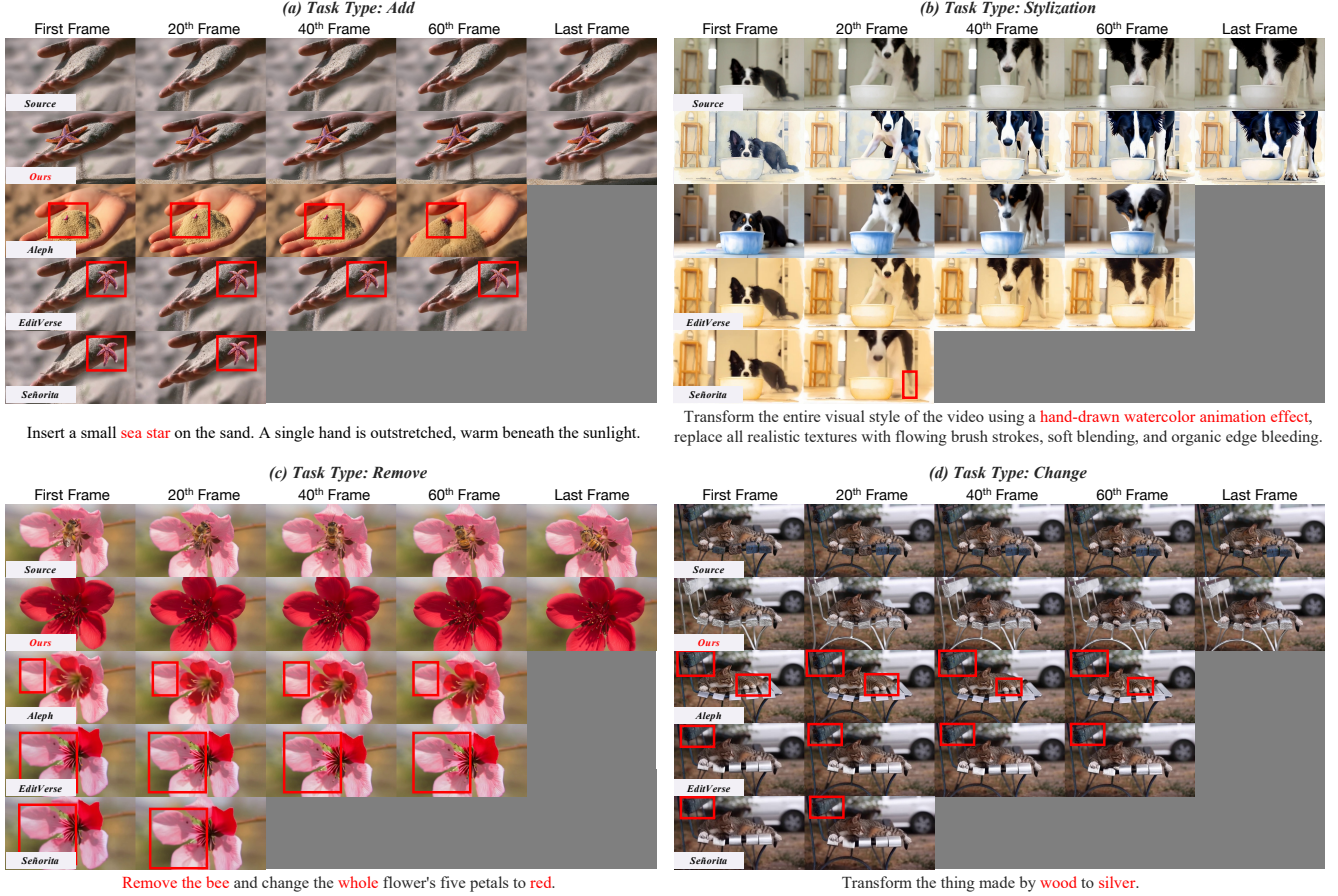


Figure 4. **Qualitative comparison.** We Choose top three method in quantitative comparison to compare with our visual results across four representative video editing tasks. Red boxes highlight the unreasonable generated contents. The gray placeholder denotes these methods cannot generate such long videos. Our method generally enjoys better editing fidelity, temporal consistency and visual quality.

Methods	EA $\uparrow$	MA $\uparrow$	VQ $\uparrow$
EditVerse [12]	4.063	3.792	3.354
Señorita-2M [41]	3.563	3.208	2.354
Aleph [1]	3.412	3.271	3.459
Ours	4.250	4.333	4.146

Table 2. User study preference regarding editing accuracy (EA), motion accuracy (MA) and video quality (VQ). Our method is consistently preferred.

Method	Temporal Consistency		Text Alignment		Video Quality Pick Score $\uparrow$	VLM Evaluation VLM Score $\uparrow$
	CLIP $\uparrow$	DINO $\uparrow$	Frame $\uparrow$	Video $\uparrow$		
Baseline	0.986	0.984	27.420	24.960	20.010	7.210
+AST-RoPE	0.989	0.988	28.178	25.817	20.354	7.542
Full	0.991	0.991	28.316	25.925	20.405	7.600

Table 3. Quantitative results for ablation variants of our model.

5.5. Ablation Study

To validate the efficacy of each component in our framework, we conduct ablation study on three variants trained with 81-frame videos: (1) **Baseline**: the original Wan-Fun model fine-tuned on our dataset without any modification, (2) **+AST-RoPE**: applying our spatial-temporal RoPE adaptation to enhance attention modules, (3) **Full**: integrating both RoPE adaptation and our proposed self-distillation

strategy. Results are summarized in Tab. 3. Thanks to our proposed dataset, the baseline model can already achieve strong performance. Based on that, the RoPE adaptation and self-distillation can further enhance the quality in terms of both visual quality and text alignment, indicating the effectiveness of the proposed method.

6. Conclusion

We addressed a core limitation in First-Frame Propagation (FFP) video editing: a foundational data gap that necessitates complicated run-time guidance which results in limited generalization ability, for which our solution covers two main aspects. First, we introduce **FFP-300K**, a large-scale dataset with high-quality and diverse videos. Second, our model leverages a novel Adaptive Spatio-Temporal RoPE (AST-RoPE) and self-distillation to strengthen the first-frame reference and source motion preservation. This dual approach achieves state-of-the-art fidelity and temporal coherence. By tackling both data and model, we make high-fidelity, controllable video editing a practical reality.